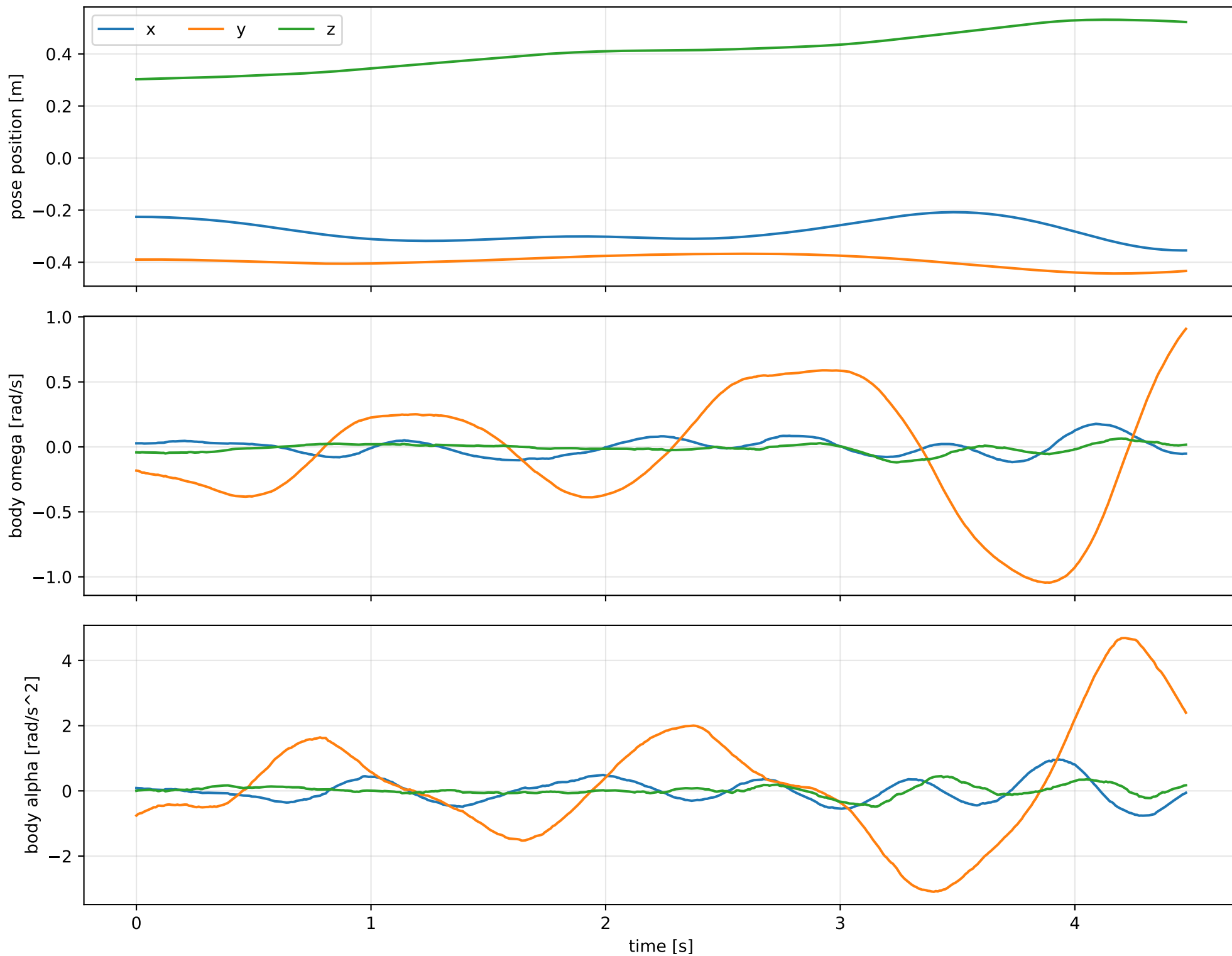
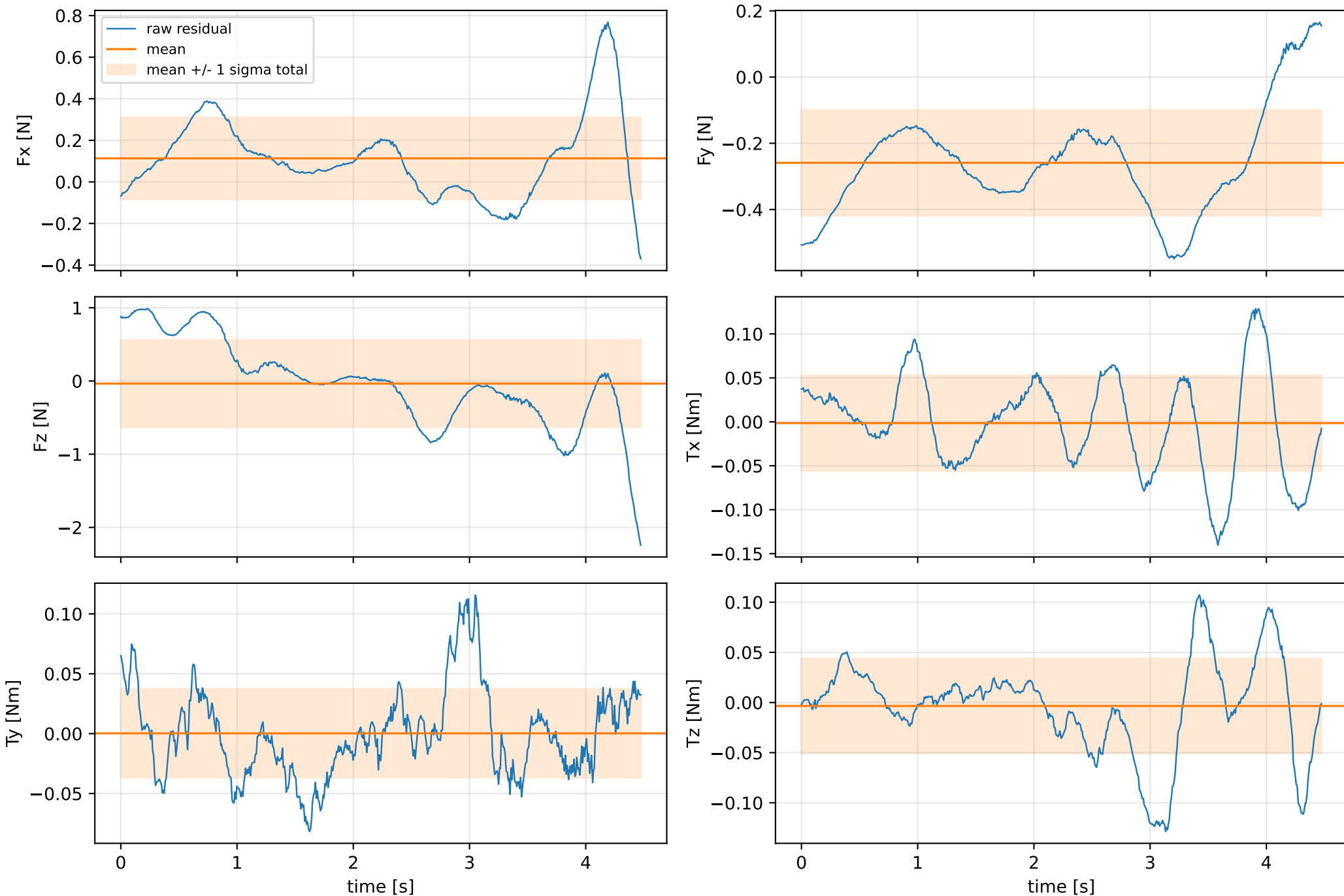


default: SG trajectory and derived motion



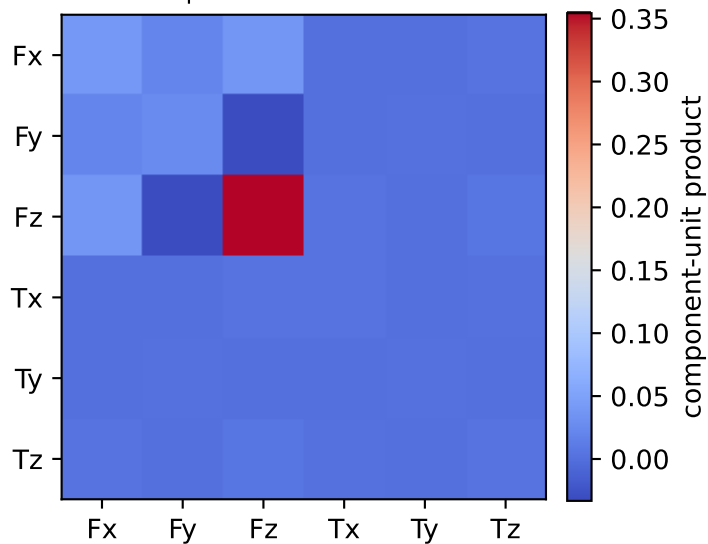
default: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



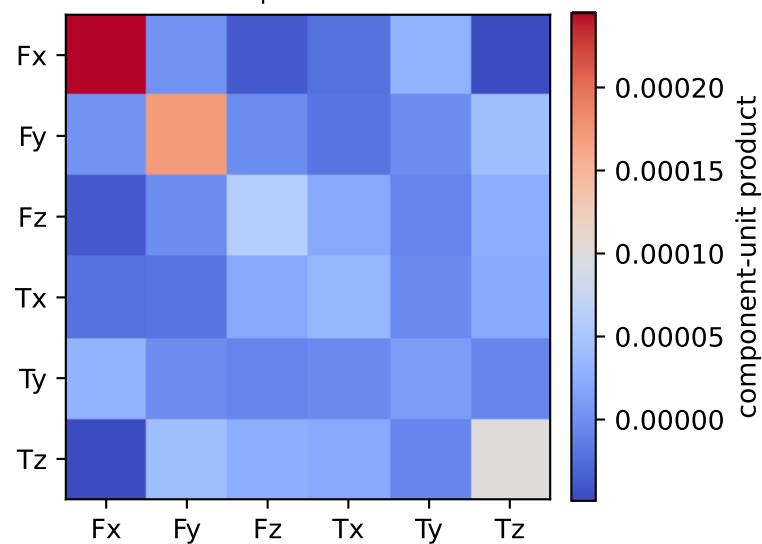
vector RMS about zero: force 0.70661 N, torque 0.080986 Nm; component std about mean: force [0.1974 0.16 0.5953], torque [0.0543 0.0371 0.0473]

default: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.002 0.004 0.005 0.052 0.361]

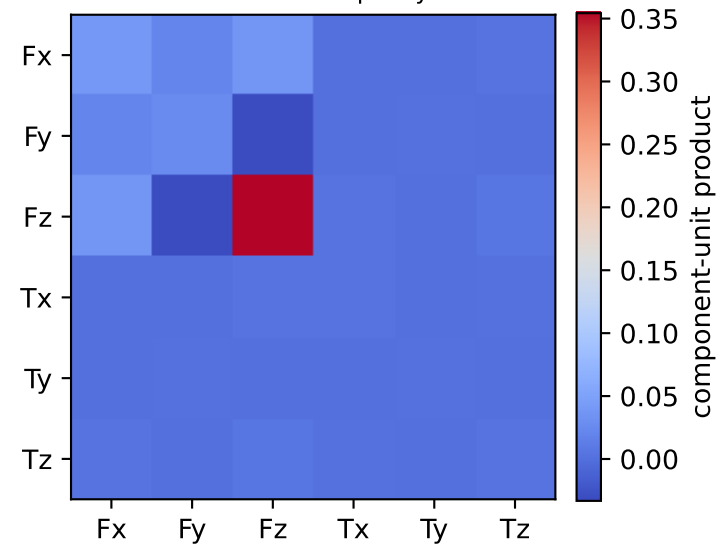
empirical total covariance



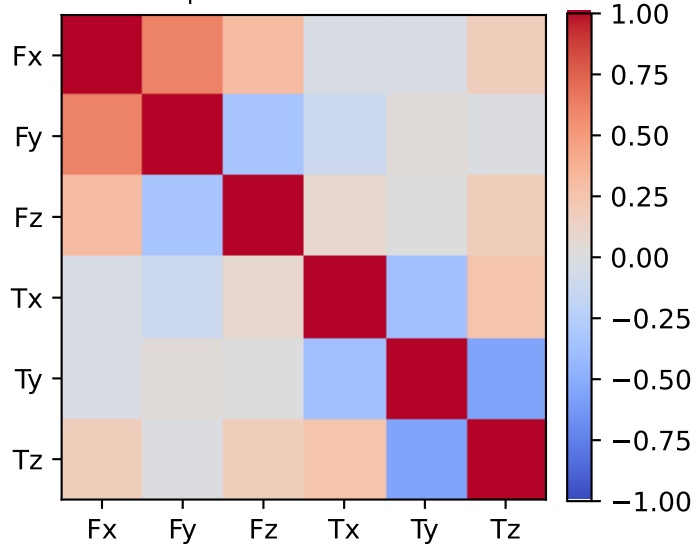
mean full-SG prediction covariance



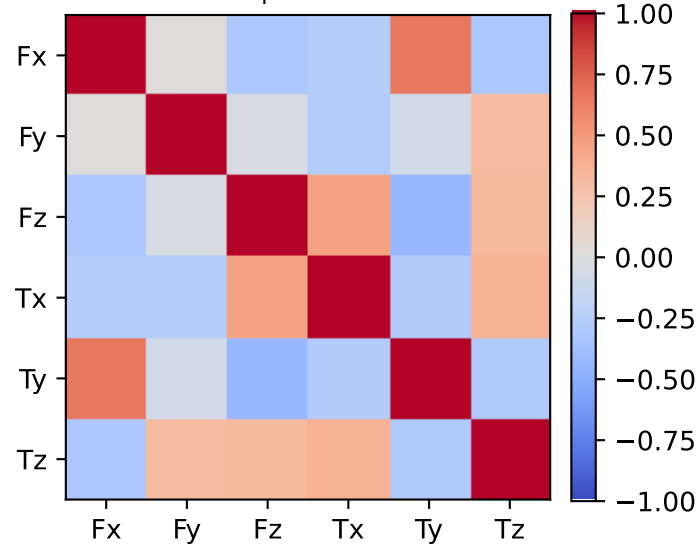
PSD excess model discrepancy covariance



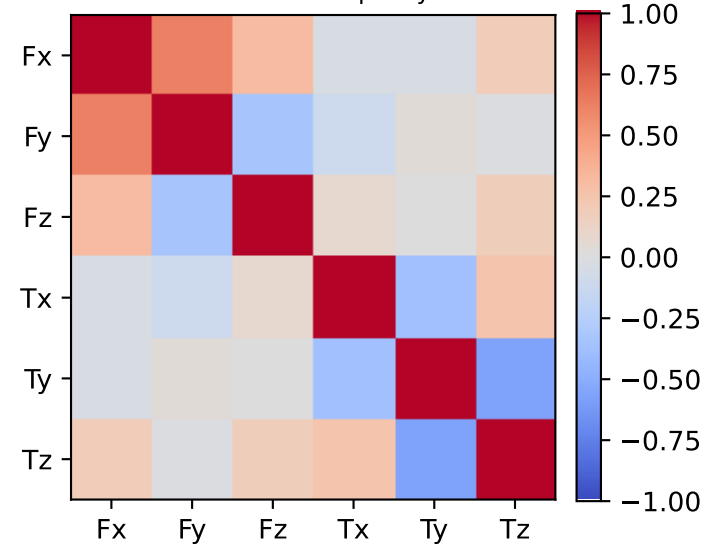
empirical total correlation



mean full-SG prediction correlation

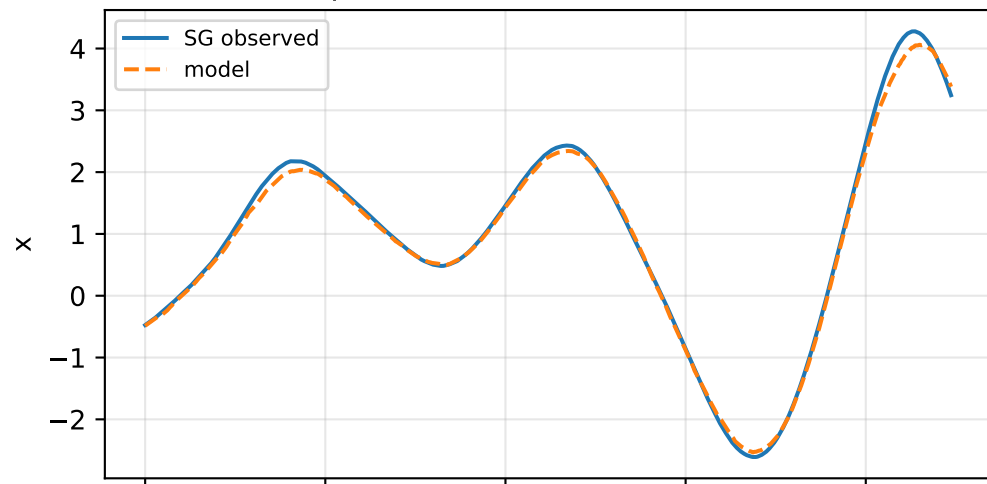


PSD excess model discrepancy correlation

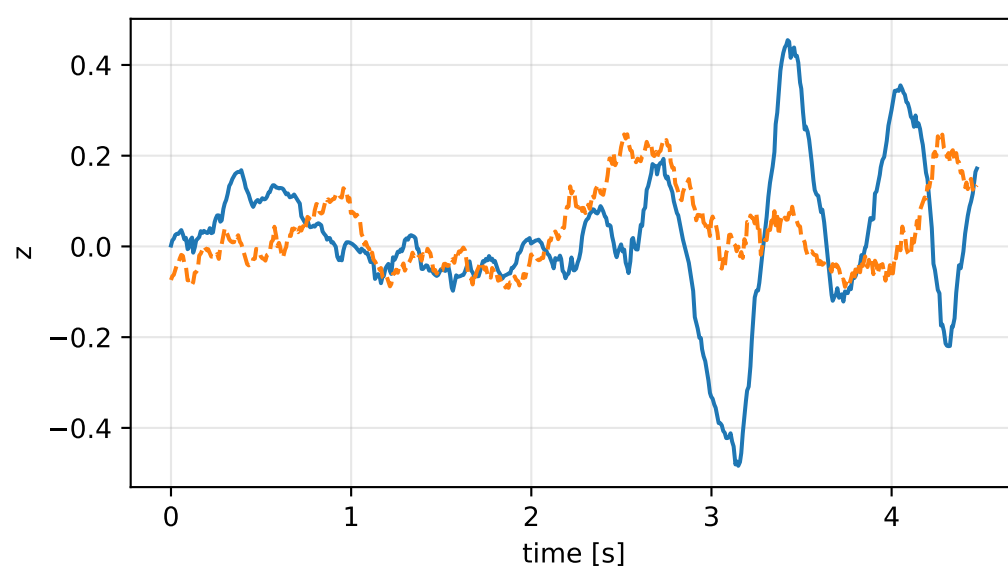
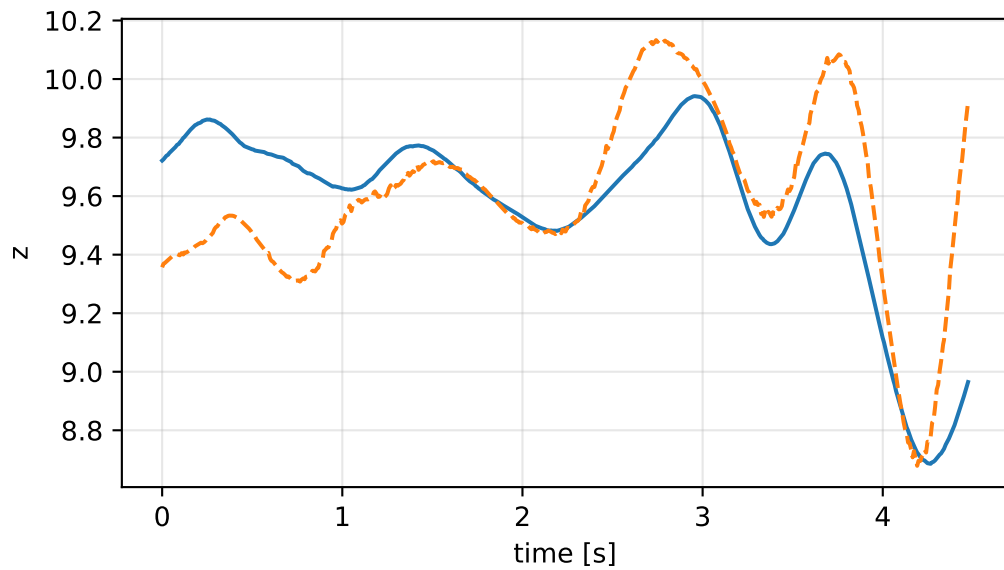
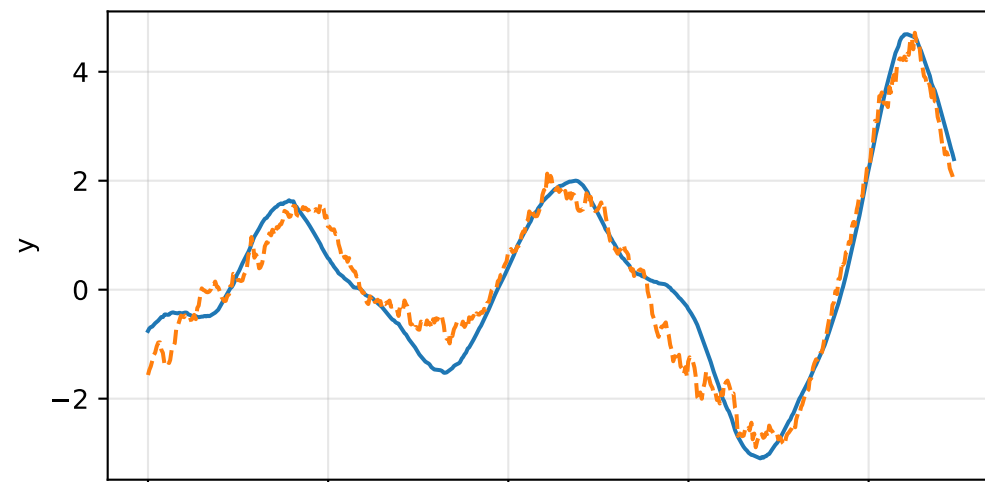
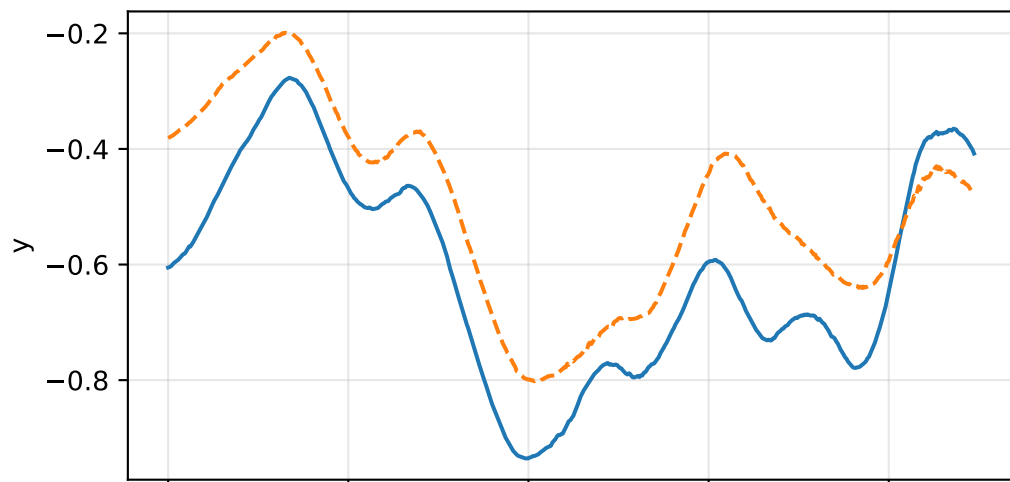
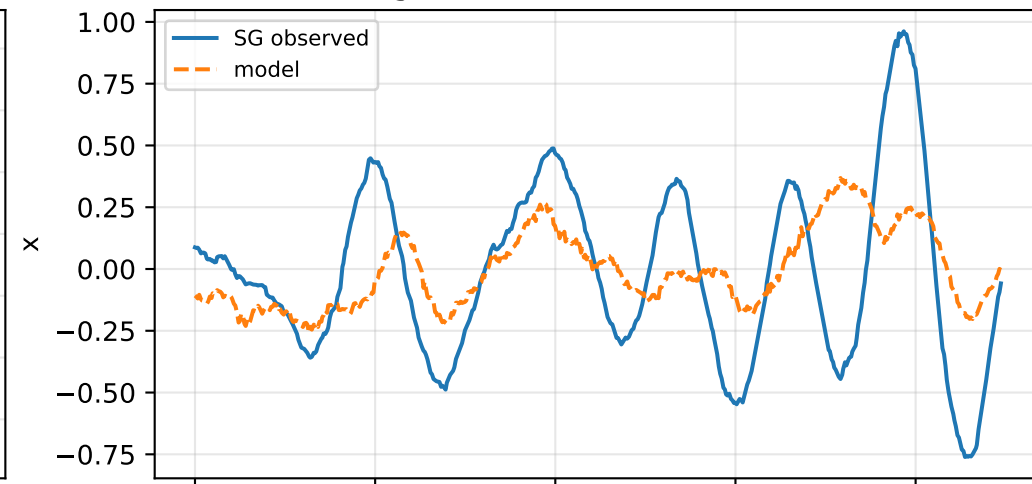


default: acceleration residual objective

specific acceleration [m/s²]

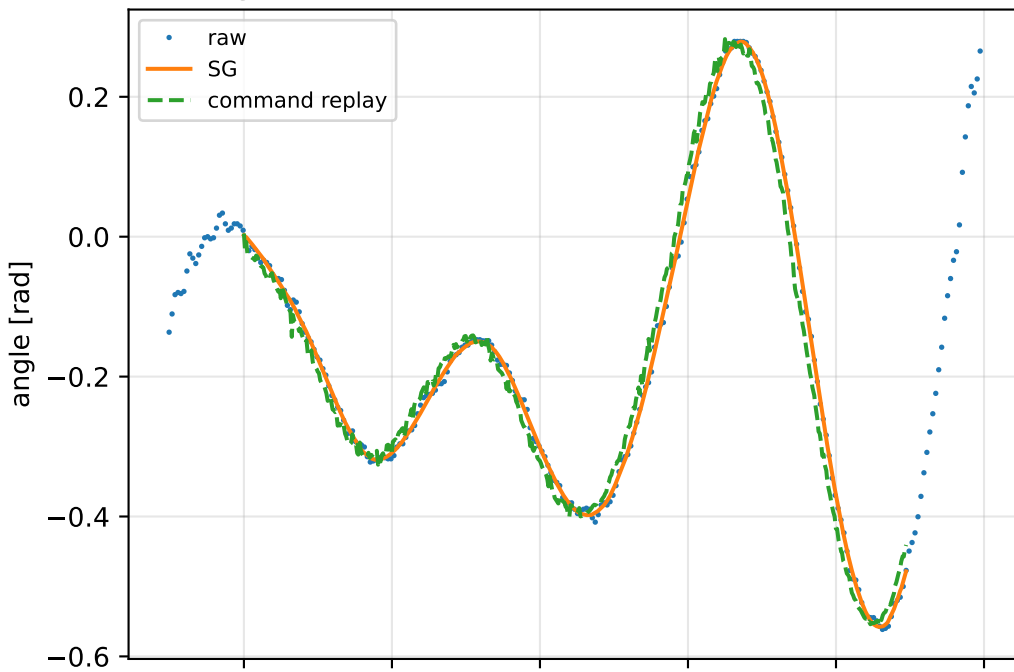


angular acceleration [rad/s²]

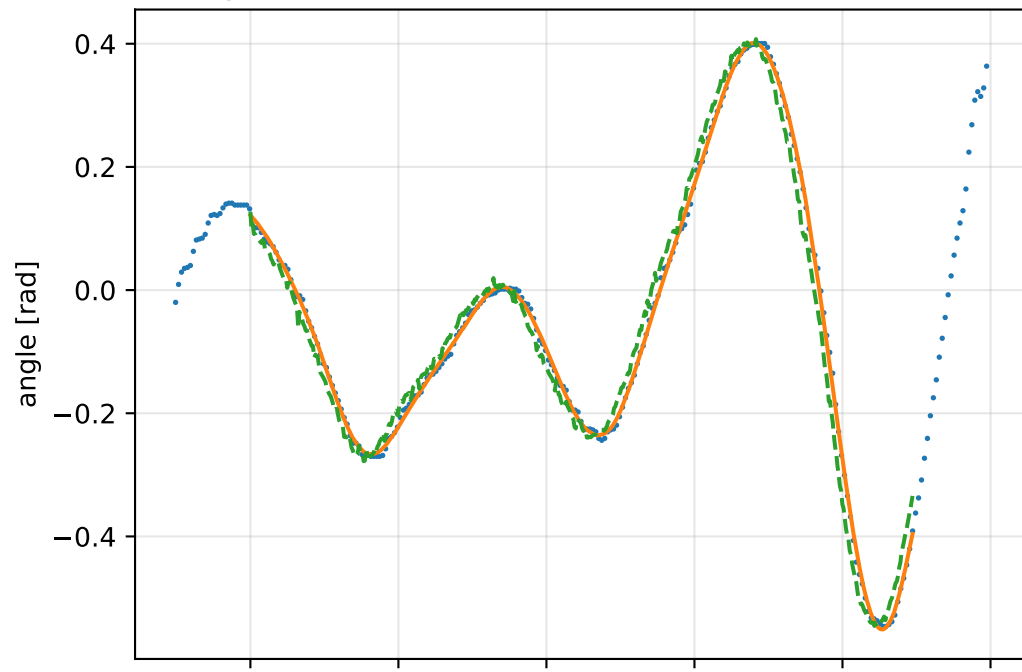


default: gimbal measurement audit (objective=measured_sg)

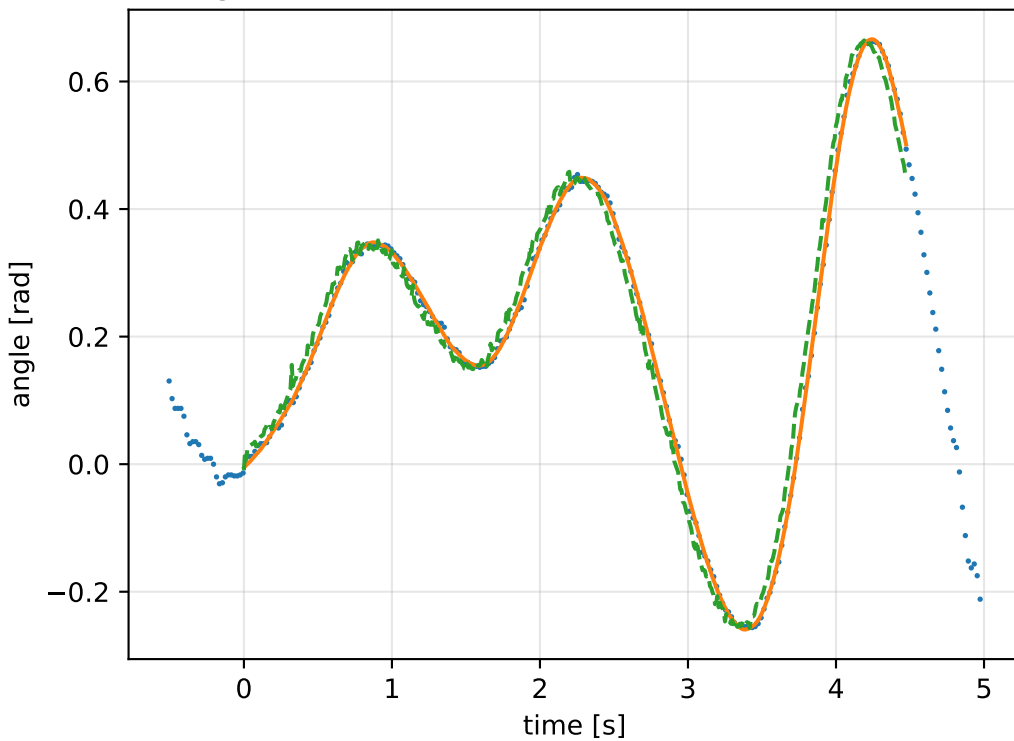
gimbal 1: RMSE=0.027, max=0.07529 rad



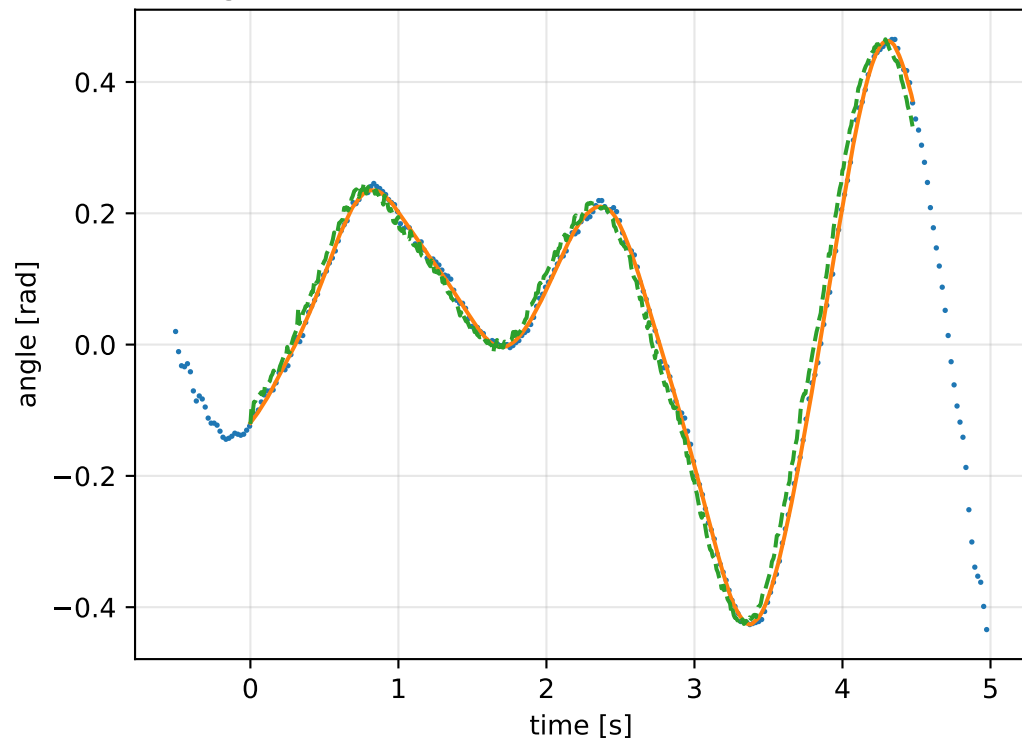
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

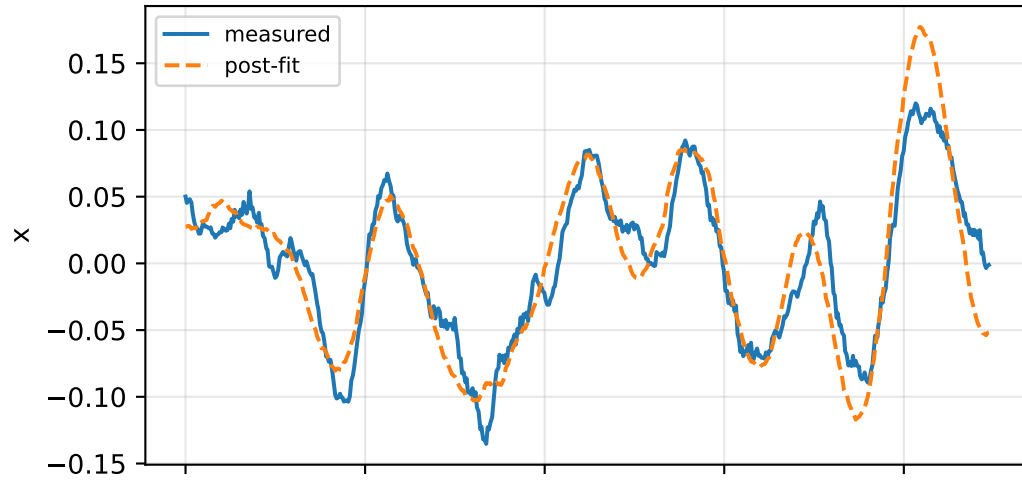


gimbal 4: RMSE=0.02598, max=0.07261 rad

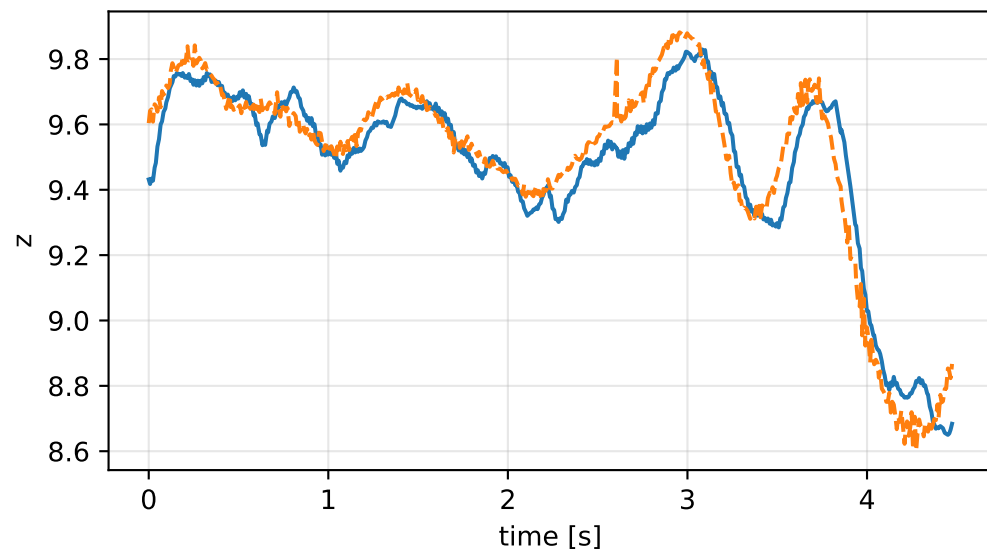
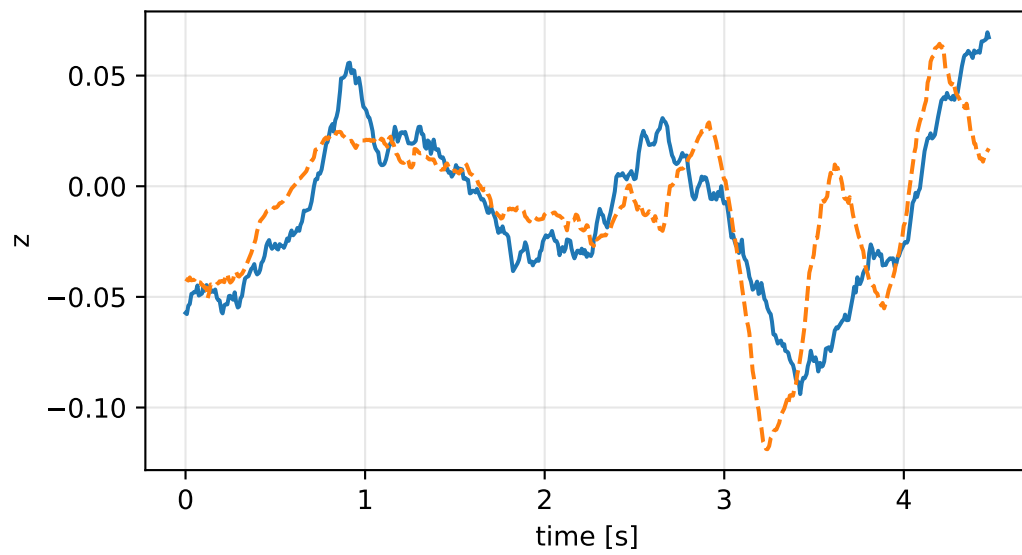
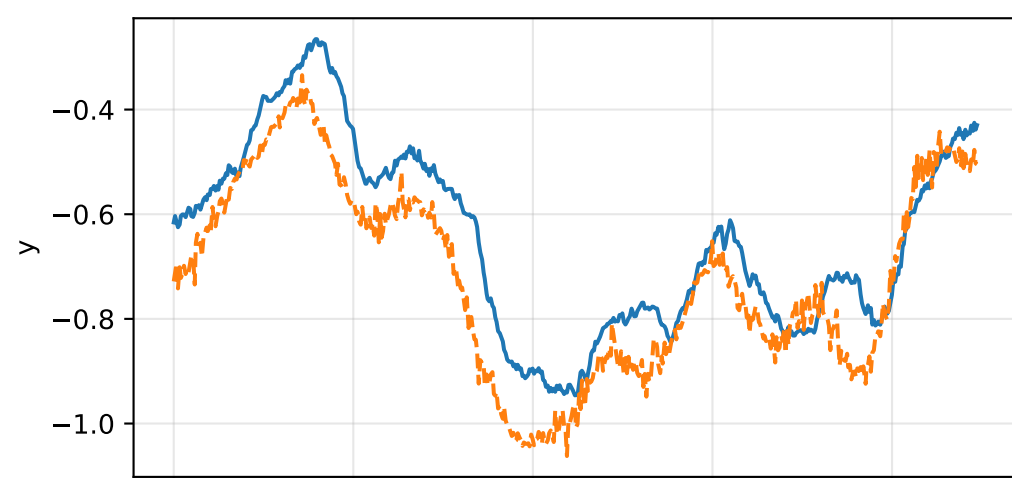
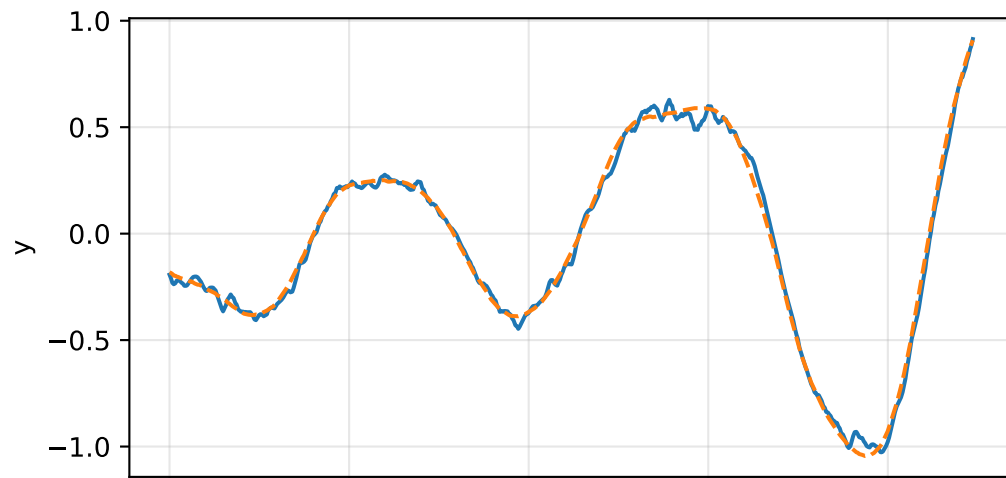
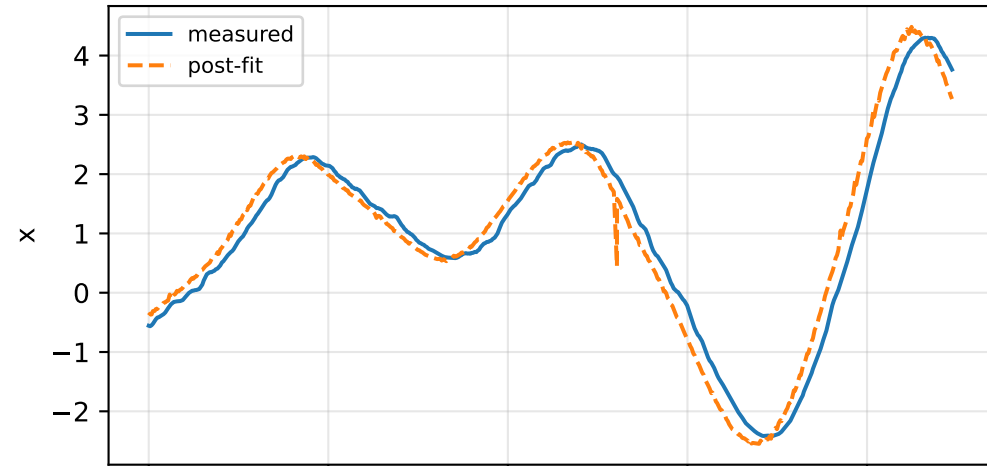


default: IMU comparison (diagnostic only)

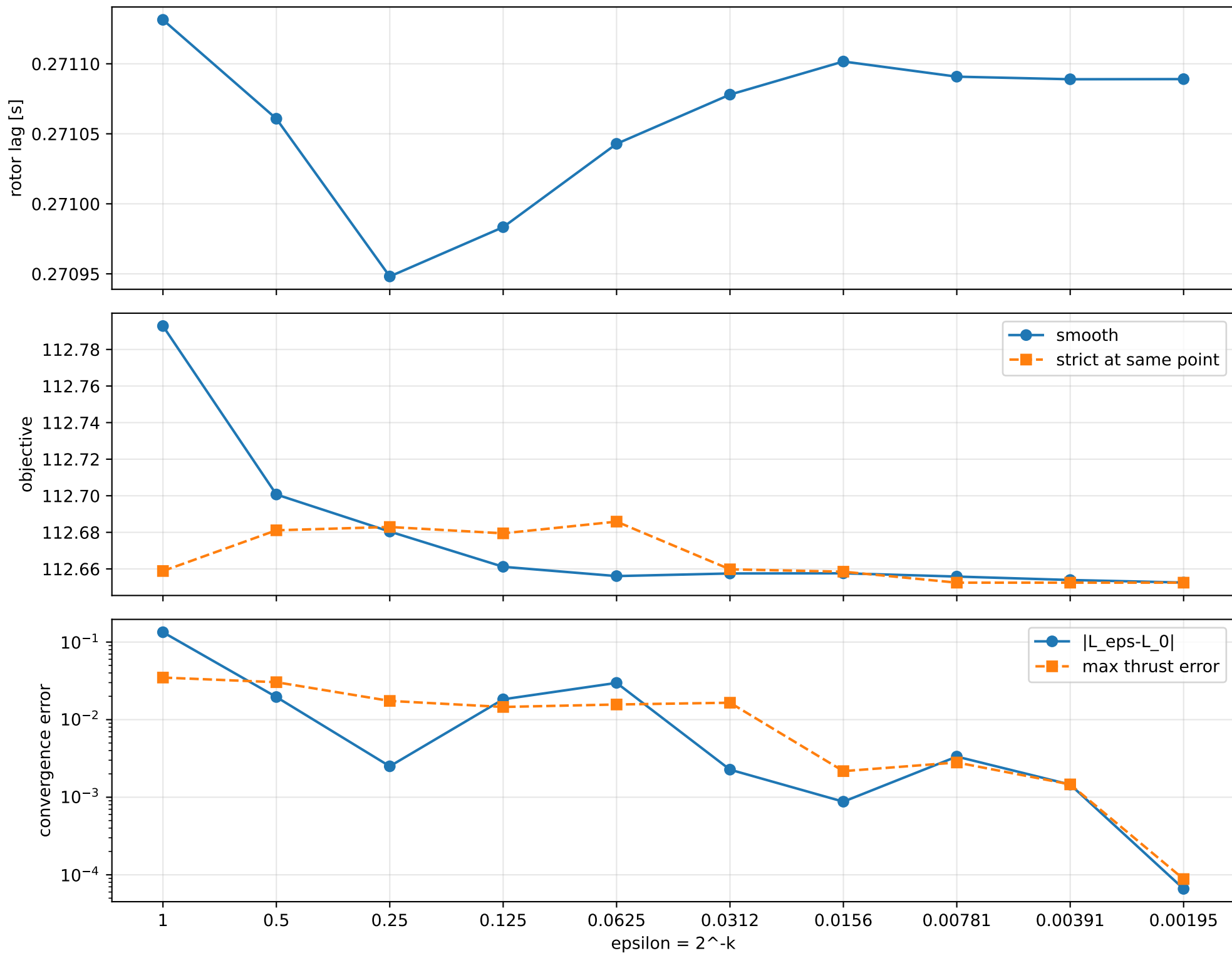
gyro [rad/s]



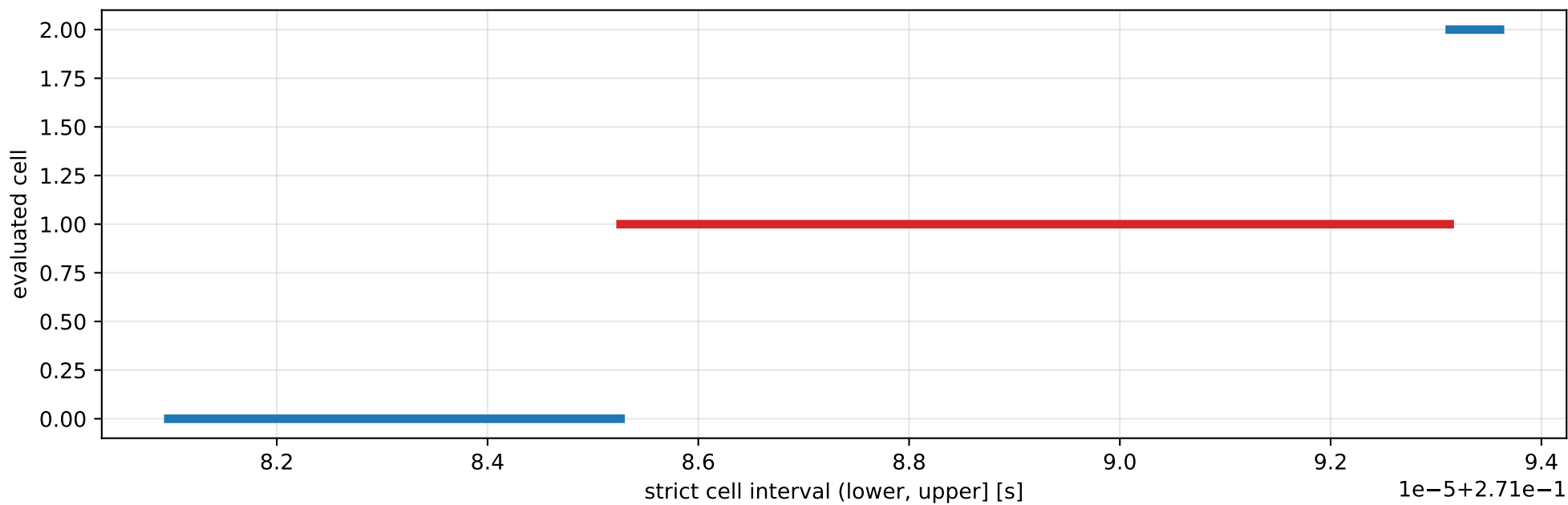
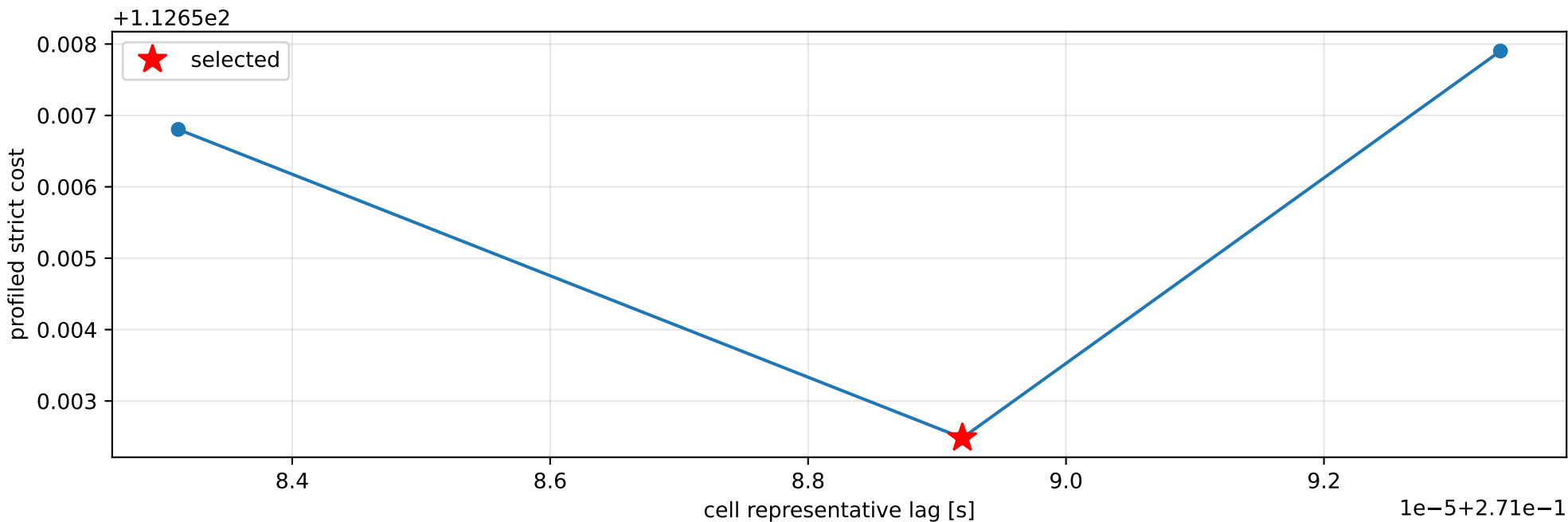
specific force [m/s²]



default: smooth-to-strict continuation



default: exact strict-ZOH lag cells



selected (0.271085262, 0.27109313] s; final smooth 0.271089061 s; data boundary=False

default: parameter estimate

Case: default

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 7.5204307

inertia [kg m²]:

```
[[ 0.55443699  0.01372501 -0.00323582]
 [ 0.01372501  0.2616833  -0.08172243]
 [-0.00323582 -0.08172243  0.79124992]]
```

CoG body [m]: [-0.026435 0.01133929 0.00941535]

force effectiveness: [2.87131322 2.32387814 2.56341438 3.07927599]

scale-free J/m [m²]:

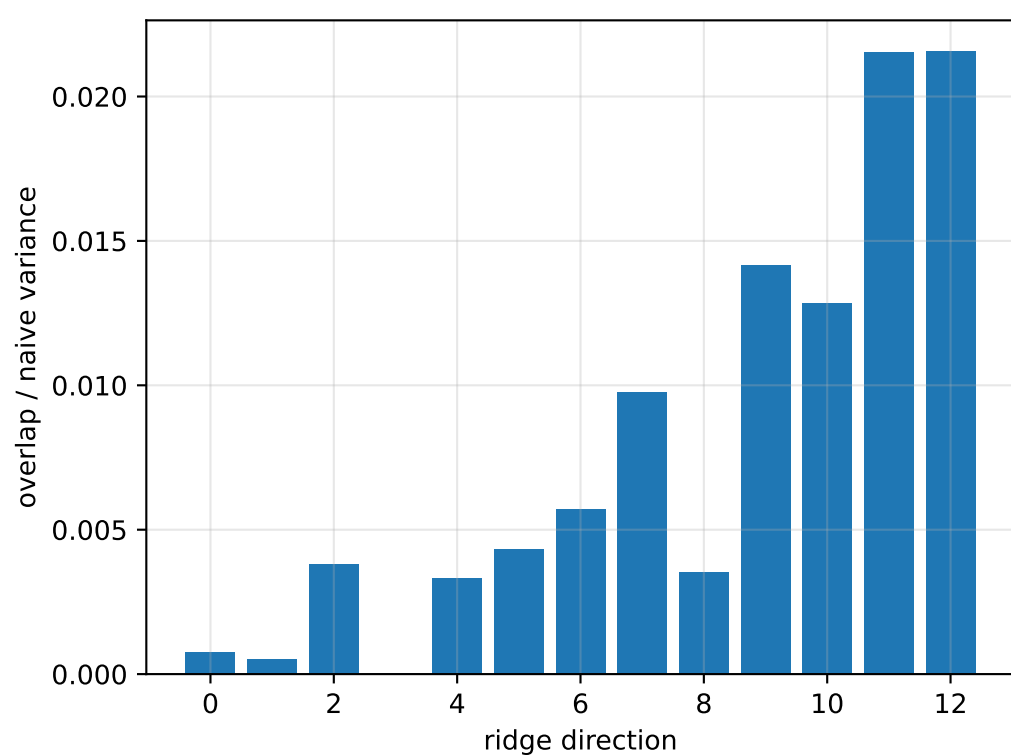
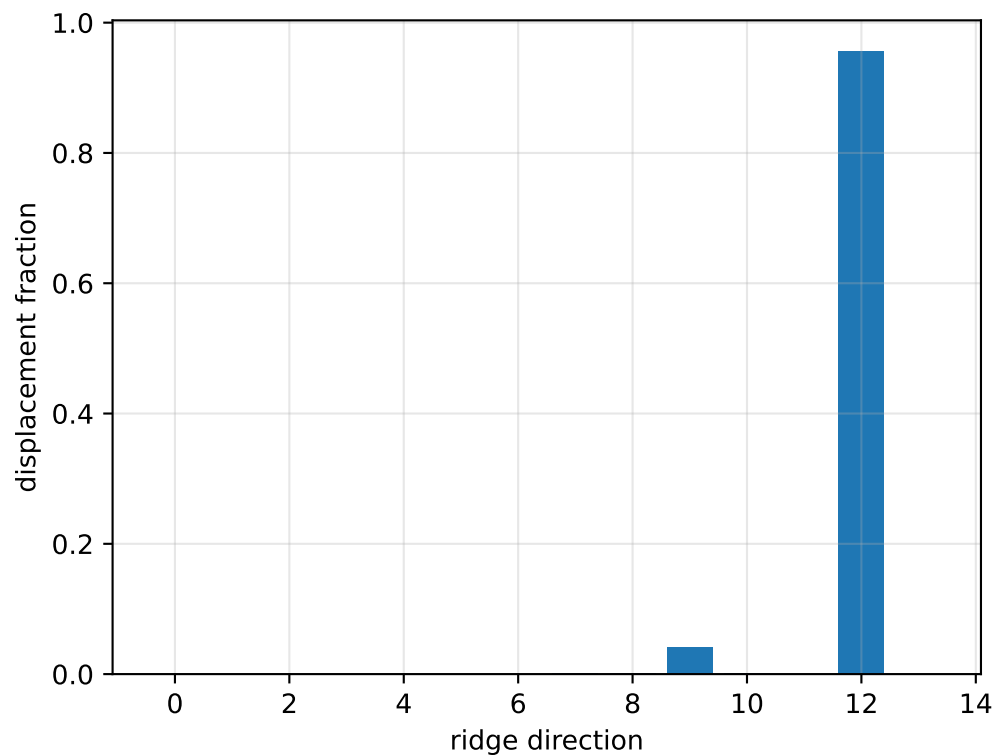
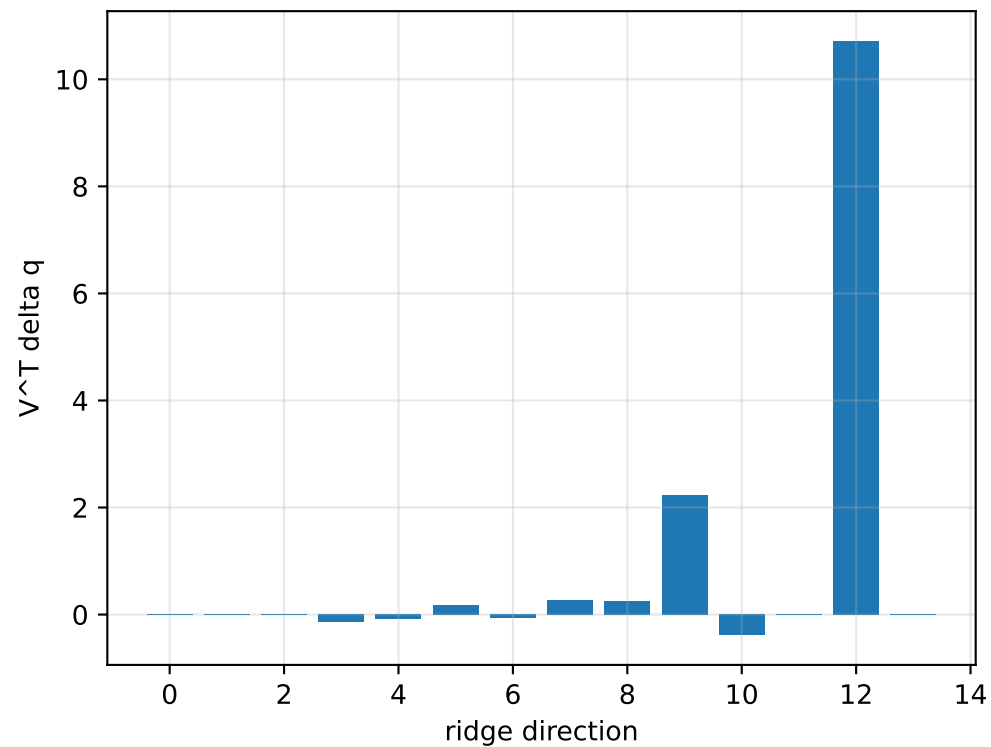
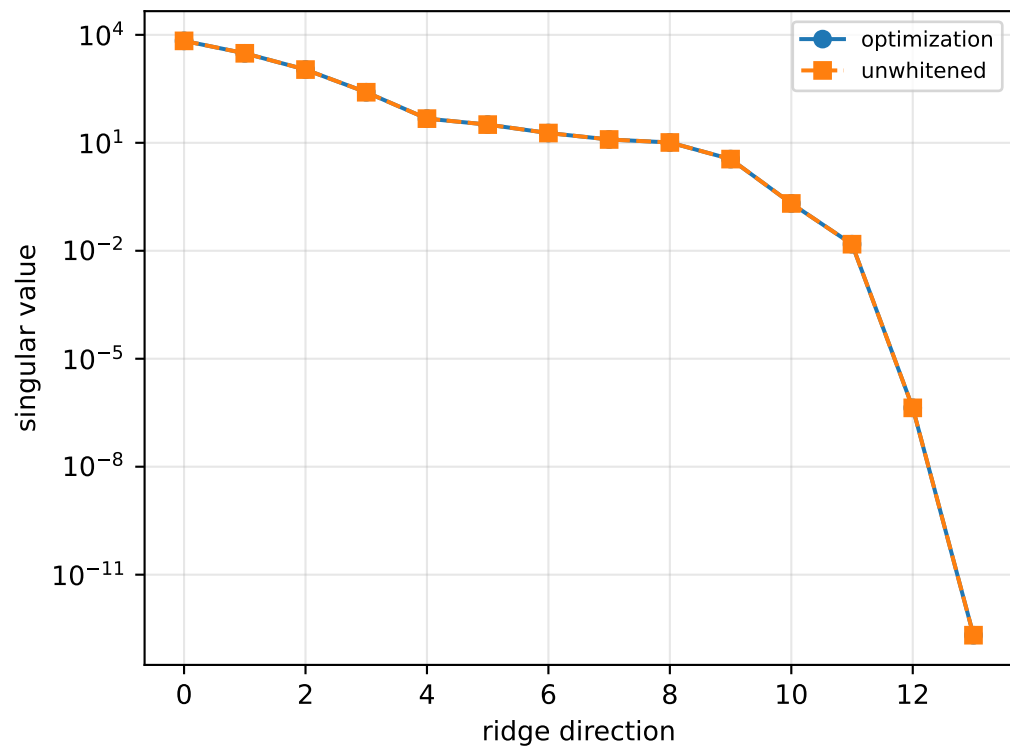
```
[[ 0.0737241  0.00182503 -0.00043027]
 [ 0.00182503  0.03479632 -0.01086672]
 [-0.00043027 -0.01086672  0.10521338]]
```

scale-free f/m: [0.3818017 0.30900865 0.34086005 0.40945474]

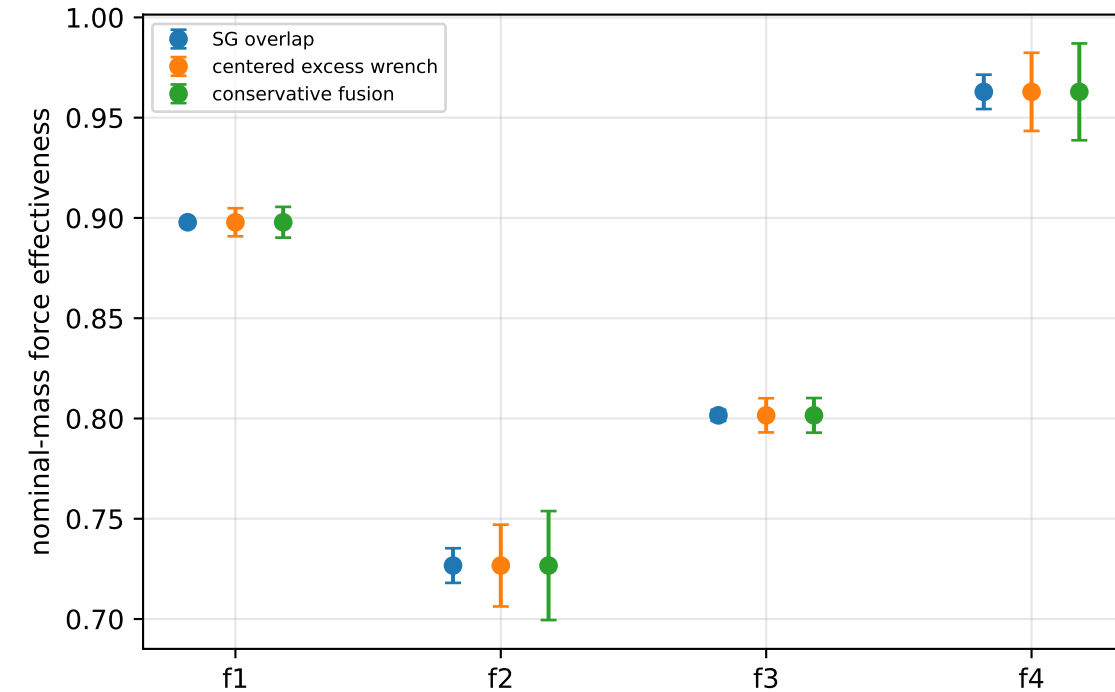
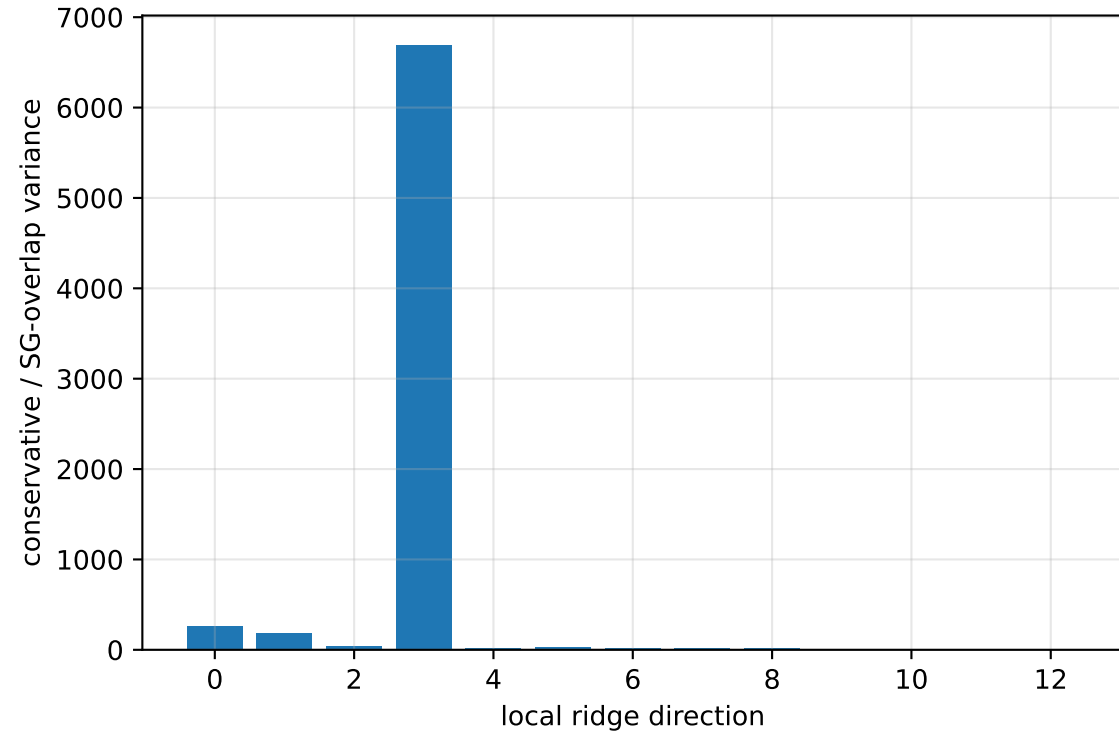
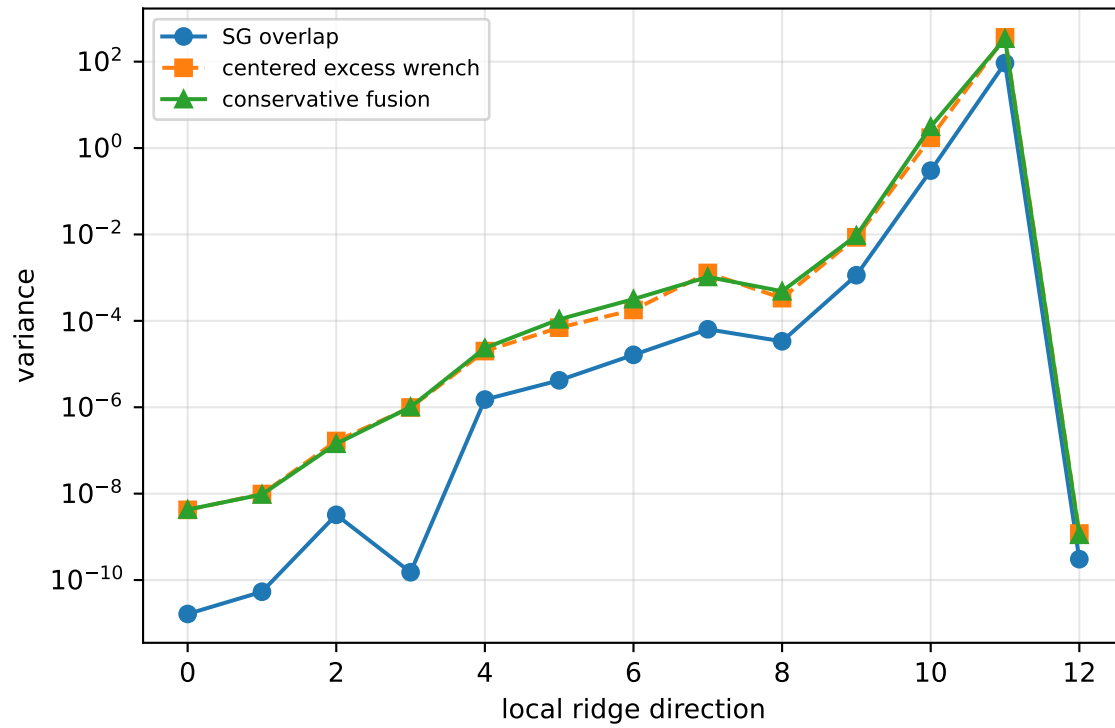
rotor lag cell: (0.271085262, 0.27109313] s

representative: 0.271089196 s

default: ridge spectrum (rank 13, nullity 1, $||\mathbf{J}\mathbf{v}||=1.62\text{e-}13$)



default: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 229.4$
wrench-to-acceleration recovery max error = $3.39\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

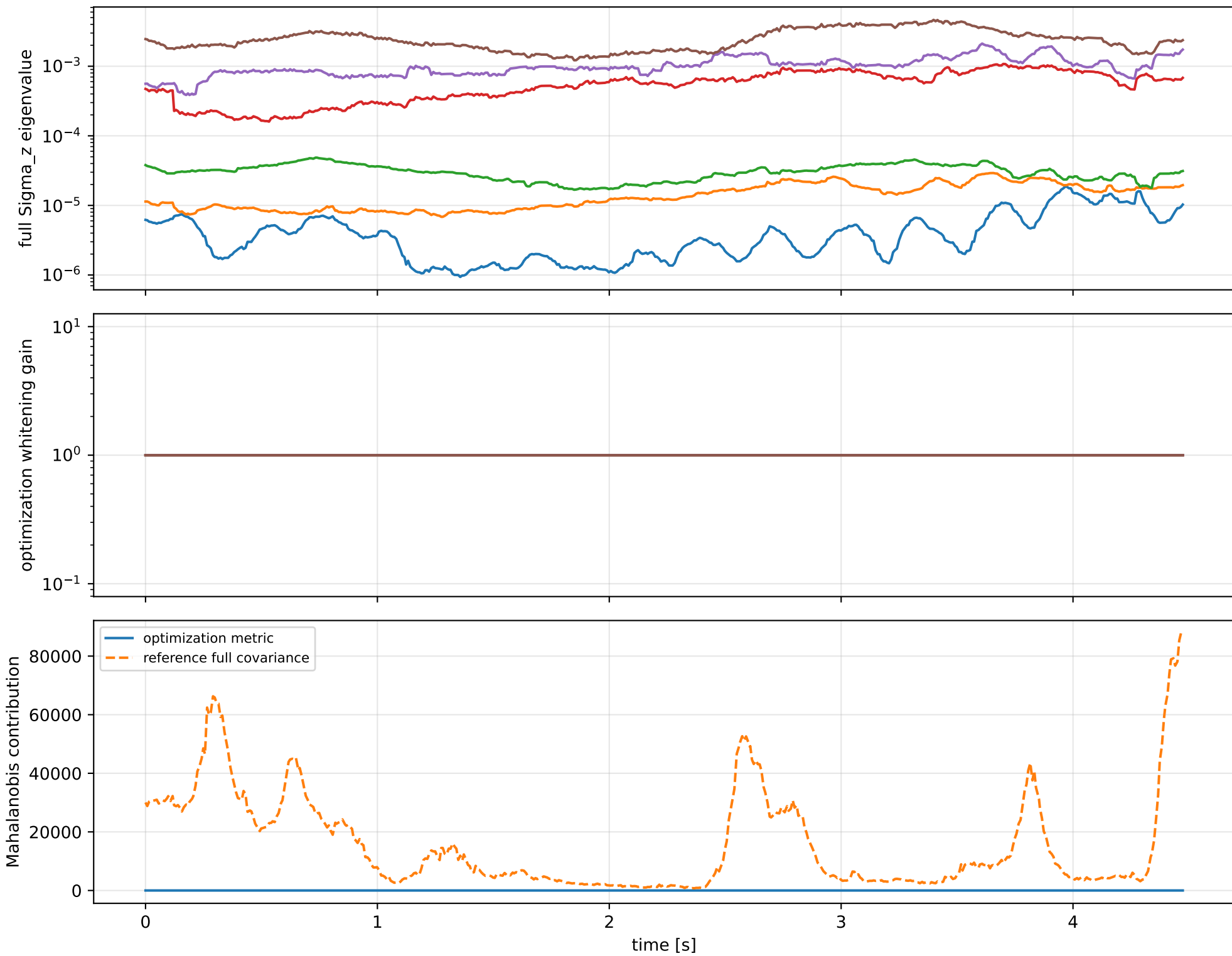
index 11: variance=340.5, ratio=3.668
 $\log_{\text{mass}}=+0.00219$; second_moment_diag_1=+0.162
second_moment_diag_2=-0.119; second_moment_diag_3=-0.0535
second_moment_offdiag_12=+0.822; second_moment_offdiag_13=-0.517
second_moment_offdiag_23=+0.114; cog_x=-5.2e-05
cog_y=+3.09e-05; cog_z=-7.22e-06
log_force_eff_1=+0.00226; log_force_eff_2=+0.00175
log_force_eff_3=+0.00208; log_force_eff_4=+0.00251

index 10: variance=3.11, ratio=10.36
 $\log_{\text{mass}}=+0.0249$; second_moment_diag_1=+0.0251
second_moment_diag_2=+0.561; second_moment_diag_3=-0.713
second_moment_offdiag_12=+0.0341; second_moment_offdiag_13=+0.0961
second_moment_offdiag_23=+0.403; cog_x=+0.000943
cog_y=-0.00131; cog_z=+0.000291
log_force_eff_1=+0.0256; log_force_eff_2=+0.0377
log_force_eff_3=+0.0224; log_force_eff_4=+0.0172

index 9: variance=0.009426, ratio=8.286
 $\log_{\text{mass}}=-0.141$; second_moment_diag_1=-0.132
second_moment_diag_2=+0.323; second_moment_diag_3=+0.567
second_moment_offdiag_12=-0.00826; second_moment_offdiag_13=-0.0513
second_moment_offdiag_23=+0.631; cog_x=-0.0198
cog_y=+0.0106; cog_z=-0.00278
log_force_eff_1=-0.104; log_force_eff_2=-0.3
log_force_eff_3=-0.188; log_force_eff_4=-0.0253

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

default: optimization vs reference covariance



default: wrench / closure (force max $2.53\text{e-}14$, torque max $2.78\text{e-}16$)

